

Krishna Gupta

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PROFESSIONAL SUMMARY

Innovative Software & AI/ML Engineer and B.Tech in Data Science student (Class of 2027) with dual industry internship experience architecting scalable distributed systems, Large Language Model (LLM) post-training workflows, and high-throughput data microservices. **Possesses a solid foundation in core system design and software architecture principles** (High-Level/Low-Level Design, microservices vs. monoliths, caching hierarchies, load balancing, database sharding & indexing, CAP theorem, and event-driven patterns). Proven expertise across the MLflow production lifecycle (tracing, evaluation, prompt templates, AI gateways, agent servers), full-stack microservices with 99.9% uptime, and certified AWS Developer skills in FastAPI, PyTorch, and relational/NoSQL optimization, delivering resilient, production-grade systems with measurable business impact.

EDUCATION

Oriental Institute of Science and Technology

Bachelor of Technology (B.Tech) in Data Science (Class of 2027)

Bhopal, Madhya Pradesh, India

Aug 2023 – May 2027

TECHNICAL SKILLS

System Design & Architecture: Core System Design Principles, High-Level Design (HLD), Low-Level Design (LLD), Client-Server & Microservices Architecture, Monolith vs. Distributed Systems, Scalability & High Availability, Load Balancing (Round-Robin, Least Connection), Reverse Proxies & API Gateways (Nginx), Distributed Caching Strategies (Redis, In-Memory, Cache-Aside), Database Indexing, Partitioning & Sharding, CAP Theorem & ACID Transactions, Asynchronous & Event-Driven Architecture, Message Queues, Rate Limiting, Fault Tolerance, Resiliency & Graceful Degradation

Generative AI & Agentic Systems: Large Language Models (LLMs), Vision-Language Models (VLMs), LangGraph, LangChain, Autonomous Multi-Agents, Hybrid Retrieval-Augmented Generation (Hybrid RAG), Prompt Engineering, Safety Guardrails, Model Alignment (RLHF/DPO), Vector Databases (FAISS, pgvector), Hugging Face Transformers

MLOps & Cloud Infrastructure: MLflow (Tracing, Model Evaluation, Prompt Templates, AI Gateways, Agent Server, Hyperparameter Tuning), Experiment Tracking, Model Registry, CI/CD Pipelines, Docker Containers, AWS Cloud Infrastructure (EC2, S3, IAM, CloudWatch)

Backend, APIs & Databases: Python (Advanced), Java, SQL (PostgreSQL, MySQL), FastAPI, RESTful APIs, Microservices, Node.js, React.js, ETL/ELT Pipelines, Dimensional Modeling (Star/Snowflake Schemas), Git, Linux Server Administration, Agile/SDLC

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, Classification, Regression, Time-Series Forecasting (LSTM), Quantitative Financial Modeling, Computer Vision (OpenCV, EasyOCR), Deep Neural Networks (CNNs), PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

WORK EXPERIENCE

Ethara.AI

LLM & AI Post Training Intern (Paid Internship)

Remote

Feb 2026 – May 2026

- Benchmarked and evaluated 50,000+ Large Language Model outputs within LLMOps workflows, boosting model alignment, precision, and safety guardrails by 15% across diverse prompt distributions.
- Designed and automated scalable Python data preprocessing, feature extraction, and validation pipelines, saving 20+ hours per week in manual quality assurance and dataset screening tasks.
- Formulated structured technical capability matrices and evaluation rubrics to ensure 100% compliance with enterprise data security, architectural standards, and privacy requirements.
- Collaborated with AI research teams to analyze edge cases, debug reward-signal discrepancies, and streamline dataset curation workflows for downstream reinforcement learning environments.

Kanchan Pvt Ltd – Web Development Wing (Sapphire)

Full Stack Development Intern (Paid Internship)

Remote

Oct 2025 – Feb 2026

- Understood and applied core system design principles to architect and deploy full-stack web infrastructure for review analytics platform “RevU Social” (revu.social) using React.js and Node.js REST APIs, driving a 20% user increase.
- Optimized PostgreSQL and MySQL relational database schemas, query execution plans, and indexing strategies, reducing query latency and accelerating transaction processing speeds by 25%.
- Implemented secure JWT authentication protocols, API rate limiting, and deployed Docker containerized microservices on Linux cloud servers, maintaining 99.9% uptime and zero security incidents.

KEY TECHNICAL PROJECTS

KALKI 1.5 – Autonomous Multi-Agent OS & MLOps Platform <i>Technologies: Python, FastAPI, LangGraph, LangChain, LLMs, VLMs, MLflow Tracing/Evaluation, Hybrid RAG, FAISS</i> Live App: kalki.hg497kg.workers.dev GitHub: kgupta171025.github.io/KALKI-1.5 - Engineered an autonomous multi-agent operating system integrating LLMs and Vision-Language Models (VLMs) via LangGraph and LangChain to automate complex multi-step reasoning workflows with decoupled agent boundaries. - Architected a low-latency Hybrid RAG pipeline integrating dense vector embeddings and FAISS with in-memory caching mechanisms, cutting semantic search query retrieval latency by 40%. - Integrated MLflow tracing, prompt evaluation templates, and automated guardrail logging to monitor inter-agent communication, audit token consumption, and ensure deterministic execution safety.	2024 – Present
Score Vision - AI (FOG Tech) – Computer Vision & Document OCR Extraction Platform <i>Technologies: Python, OpenCV, EasyOCR, Sentence-Transformers, FAISS, FastAPI, React, Docker</i> Live App: scorevision-ai.hg497kg.workers.dev GitHub: kgupta171025.github.io/FOG_Technologies - Developed an intelligent OCR and computer vision pipeline achieving 92% text extraction accuracy and sub-200ms semantic matching on complex, unstructured imagery. - Designed a decoupled client-server microservice architecture using FastAPI and Docker to handle high-throughput image processing and FAISS vector similarity search at scale.	2025 – Present
SHELF-SCANNER – High-Speed Computer Vision & Document OCR Platform <i>Technologies: Python, OpenCV, EasyOCR, Sentence-Transformers, FAISS, FastAPI, Cloudflare Workers</i> Live App: shelf-scanner.hg497kg.workers.dev GitHub: kgupta171025.github.io/SHELF-SCANNER Engineered an image-to-text OCR and semantic book-matching platform using EasyOCR and FAISS vector search, achieving sub-200ms query retrieval latency.	2025 – Present
Future Stock Market Prediction & Quantitative Financial Modeling <i>Technologies: Python, Scikit-learn, PyTorch MLOps, MLflow Hyperparameter Tuning, Time-Series (LSTM)</i> Live App: smf-ai.web.app GitHub: github.com/KGupta171025 - Built a quantitative time-series forecasting model using PyTorch (LSTM) and Scikit-learn to predict financial asset price trends with 88% precision. - Implemented MLflow hyperparameter tuning and experiment tracking to systematically evaluate learning rates, sequence lengths, and optimizer parameters.	2024 – 2025
NITI AI – Multilingual Scheme Intelligence & Financial Incentive Platform <i>Technologies: Next.js 14, React 18, TypeScript, Tailwind CSS, Cloudflare Workers, Firebase Auth, Cloud Firestore</i> Live App: niti-ai.web.app GitHub: github.com/KGupta171025/NITI-AI - Architected a multilingual AI scheme intelligence platform evaluating demographic eligibility, enterprise sector, and funding needs against 100+ national and state schemes. - Engineered serverless edge deployments on Cloudflare Workers and Firebase with static asset streaming and client-side caching, achieving sub-100ms response times.	2025 – Present
RevU Social – Full-Stack Review & Analytics Platform <i>Technologies: React.js, Node.js, REST APIs, MySQL, PostgreSQL, Star Schema revu.social</i> Designed dimensional star-schema database models in MySQL/PostgreSQL and architected modular API endpoints to power real-time business intelligence and performance dashboards.	Oct 2025 – Feb 2026
Razorpay Recovery – Smart Payment Recovery & FinTech Transaction Pipeline <i>Technologies: Python, SQL, PostgreSQL, REST APIs, Webhooks, Data Pipelines, Transaction Analytics</i> Live App: razor-recovery.hg497kg.workers.dev GitHub: kgupta171025.github.io/Razor-Recovery - Constructed an automated, event-driven transaction recovery pipeline capturing failed payment webhooks in real time with idempotent processing, successfully recovering 18% of abandoned checkouts. - Designed analytical star-schema databases in PostgreSQL to aggregate transaction failure patterns, automate notification triggers, and generate recovery analytics.	2025

CERTIFICATIONS

• AWS Certified Developer Associate (Infosys, Jun 2026)	• Machine Learning with Python (IBM SkillsBuild, Jun 2026)
• Data Science & Analytics (HP LIFE, Jun 2026)	• Neural Networks & CNNs (LinkedIn, Jun 2026)
• Deloitte Australia Data Analytics Simulation (Forage, Sep 2025)	• Tata Data Visualisation Simulation (Forage, Sep 2025)

RELEVANT COURSEWORK

System Design & Architecture • Distributed Systems • Data Structures & Algorithms • Database Management Systems • Operating Systems • Computer Networks • Software Engineering • Machine Learning • Big Data Engineering